

Date: 01/06/2026

My objectives for the day:

→ To Understand the structure and Objectives of the AI Summer Internship program. To learn about ethical Considerations in AI development and deployment.

List of activities I did today:

- learn about Real-world applications of AI in healthcare, agriculture, education, finance and Autonomous systems.
- Installed and configured python and necessary development libraries
- Installed and verified the CUDA Environment for NVIDIA GPU Support
- Tested the setup by running sample python commands and verifying CUDA availability.

My critical observation:

- AI is transforming multiple industries and creating innovative solutions to complex problems
- GPU acceleration significantly improves computational performance compared to CPU-only execution.

What did I learn today:

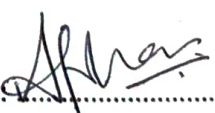
- Current trends and future opportunities in AI.
- Importance of data privacy, fairness and transparency in AI systems
- Basics of CUDA and GPU computing.

What I would have done better:

- prepared the required software and dependencies before the session to reduce setup time
- Explored additional documentation on CUDA and GPU architecture.

My plan for tomorrow:

- Review today's concepts related to AI fundamentals and ethics
- Explore and verify that the development environment is functioning correctly

Student Signature: Faculty-in-Charge Signature: 

Date: 02/06/2026

My objectives for the day:

- To Understand the fundamental concepts of Machine learning
- To learn the difference b/w Regression and Classification problems

List of activities I did today:

- Explored logistic Regression models for classification tasks such as Survival prediction and P/F Prediction.
- learned the concepts of training datasets, testing datasets, and train - test Splitting.
- Analyzed Confusion Matrices to evaluate classification Model accuracy
- learned about Gpu-Supported ML workflows and the advantages of accelerated computation.

My critical observation:

- Proper data preprocessing significantly impacts model performance.
- Gpu acceleration can reduce training time and improve computational efficiency for larger datasets.

What did I learn today:

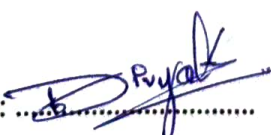
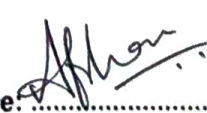
- Dataset preparation, feature selection and model training
- practical implementation of predictive models using python and scikit-learn.

What I would have done better:

- Explore additional Machine learning datasets for practice
- practiced implementing Machine learning pipelines independently

My plan for tomorrow:

- continue hands-on experimentation with datasets and performance evaluation
- learn more about model optimization techniques

Student Signature: Faculty-in-Charge Signature: 

Date: 03/06/2026

My objectives for the day:

- learn neural networks and backpropagation.
- Understand pyTorch basics on H200 GPU

List of activities I did today:

- studied neural networks from scratch
- learned backpropagation concepts
- Explored PyTorch fundamentals
- Implemented simple neural network models
- Executed training on GPU

My critical observation:

- Neural networks learn by adjusting weights through backpropagation
- GPUs significantly accelerate training

What did I learn today:

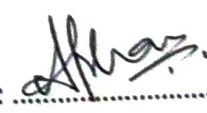
- Neural network architecture
- Backpropagation process
- Basics of pyTorch framework

What I would have done better:

- practiced more pyTorch coding examples
- Explored more advanced neural network architectures
- performed additional experiments with different learning rates

My plan for tomorrow:

- learn Convolutional Neural Networks (CNNs)
- perform image classification experiments

Student Signature: Faculty-in-Charge Signature: 

Date: 04/06/2026

My objectives for the day:

- Understand Convolutional Neural Networks
- Learn image classification techniques
- Explore GPU profiling tools

List of activities I did today:

- Attended a lecture on CNN architecture
- Learned about convolution, pooling, and feature extraction
- Studied image datasets and preprocessing techniques
- Implemented CNN models using PyTorch
- Performed image classification experiments

My critical observation:

- CNNs are highly effective for computer vision applications
- GPU profiling helps identify performance bottlenecks

What did I learn today:

- CNN architecture and working principles
- GPU profiling and optimization basics
- Importance of data preprocessing

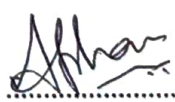
What I would have done better:

- Experimented with deeper CNN architectures
- Explored additional optimization methods

My plan for tomorrow:

- Prepare for the AI fundamentals Quiz
- Clarify doubts with mentors before moving to advanced AI topics

Student Signature: 

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Date: 05/06/2026

My objectives for the day:

- Explore Creative applications of web Technologies
- Improve problem-solving and analytical skills

List of activities I did today:

- Explored interactive web-based simulations and demonstrations.
- Learned how creativity and technology can be combined to enhance user experience.
- Engaged in discussions about innovative applications of AI and modern technologies.

My critical observation:

- Interactive technologies make learning more engaging and enjoyable.
- Continuous revision is essential for retaining technical knowledge.

What did I learn today:

- Techniques for improving problem-solving and logical thinking skills.
- Significance of revision and self-assessment in learning.

What I would have done better:

- Participated more actively in discussions and practical activities.

My plan for tomorrow:

- Learn about Transformer Architecture and Attention Mechanisms.
- Prepare for advanced AI and Natural language processing concepts.

Student Signature: Faculty-in-Charge Signature: 

Date: 08/06/2026

My objectives for the day:

- Advance AI, LLM's & Capstone Project
- Transformer Architecture deep dive / Attention mechanisms BERT & GPT Anatomy

List of activities I did today:

on day 6, I explored the architecture of transformer model which forms the best system such as BERT & GPT. In this session focused on understanding how transformer process sequential data using attention mechanism instead of traditional neural network.

My critical observation:

Self-attention help the model focus on important words in sequence. Training loss decreased over epochs, showing effective model learning.

What did I learn today:

- Input Embeddings
- Self-Attention Mechanism
- Transformer training process
- GPT style causal masking

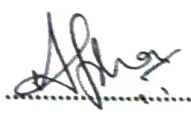
What I would have done better:

→ Explored the mathematical calculation behind attention score in more detail. Visualized attention weights to better understand token relationship.

My plan for tomorrow:

- LLM fine-tuning with LoRA & PEFT
- Hugging Face on H200
- Prompt engineering

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Date: 09/06/2026

My objectives for the day:

- LLM Fine-Tuning with LORA & PEFT
- Hugging face on H200
- prompt engineering

List of activities I did today:

- Studied the fundamentals of Large Language Model
- learned about LORA for efficient model fine-tuning
- Explored PEFT Techniques
- worked with Hugging face ecosystem & transformer model
- Practised prompt engineering techniques.
- Studied methods to optimize model performance while reducing computational cost

My critical observation:

- LORA Significantly Reduces number of trainable parameters. PEFT enables efficient fine-tuning with lower computational cost.

What did I learn today:

Prompt engineering plays vital role, it depends on how we give prompt for expecting answer.

What I would have done better:

Tested different prompt engineering greatly improve AI and response. PEFT techniques behind LORA.

My plan for tomorrow:

Generative AI: Diffusion Model & GANs
Stable Diffusion on H200 GPU

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Date: 10/06/2026

My objectives for the day:

- Generative of AI
- Diffusion Models & GAN's
- Stable Diffusion on Hw GPU

List of activities I did today:

- Attended the lecture on Generative AI concepts
- learned the difference b/w Discriminative AI and Generative AI
- Studied GANs including Generator and Discriminator architecture

My critical observation:

- GANs can generate image very quickly but are difficult to train because of instability and mode collapse issues.

What did I learn today:

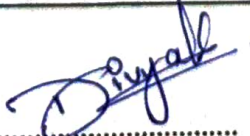
- Generative AI and differences b/w other AI
- GAN & how does it work.

What I would have done better:

- Spend more time understanding the mathematical concepts behind diffusion process

My plan for tomorrow:

- learn GPU optimization techniques such as mixed precision training

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Date: 11/10/2026

My objectives for the day:

- GPU Optimization: Mixed Precision
- Flash attention, Tensor Parallelism
- Capstone work

List of activities I did today:

- learned GPU optimization techniques like mixed precision and tensor parallelism
- Quiz from (Day 01 to Day 09)
- Practice of cv problems & video

My critical observation:

- I could not answer question (during Quiz)

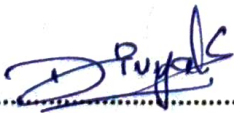
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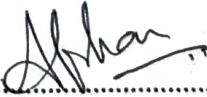
Video (helps in OpenCV)

What I would have done better:

My plan for tomorrow:

Capstone Project presentation
peer review
Certificate
Closing Ceremony

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Date: 12/06/2026

My objectives for the day:

- GPU Optimization : Mixed Precision
- Flash attention , Tensor Parallelism
- Capstone work -

List of activities I did today:

- Delivered our teams Capstone project presentation and Question from evaluation panel
- Actively participated in Peer review session

My critical observation:

I noticed that teams who focused heavily on practical implementation and solve problem solving metrics tended to handle better

What did I learn today:

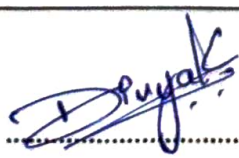
learned how to effectively structure and articulate complex technical concepts.

What I would have done better:

I could do better by adding features for our project "Group core AI"

My plan for tomorrow:

To revise all concepts, work one more project

Student Signature: Faculty-in-Charge Signature: 